Pump Sizing and Piping

UBC CHBE capstone · Montbretin A production · Media sterilization (PFD 100)

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Part A: Method

1. Continuity (basis)

$$\dot{m}_{\text{in}} = \dot{m}_{\text{out}}, \qquad \dot{m} = \rho\,\dot{Q} \quad \xrightarrow{\text{incompressible flow}} \quad \dot{Q}_{\text{in}} = \dot{Q}_{\text{out}}$$

2. Pipe sizing

$$\dot{Q} = A\,u \quad \Rightarrow \quad A_{\min} = \frac{\dot{Q}}{u_{\max}} \tag{1}$$

$\dot{Q}$ comes from the mass balance. $u_{\max}$ is a course heuristic: 3 m/s for liquids, 10 m/s for gases. $A_{\min}$ is the minimum cross-sectional area; the nominal pipe size, schedule and inner diameter D_{inner} are then read from Table 19.1 (Dimensions of Standard Steel Pipe): choose the smallest Schedule 40 size whose inside cross-sectional area exceeds $A_{\min}$, then recompute the actual velocity.

3. Total head

Motor power needs the elevation, pressure and friction heads:

$$h_{\text{total}} = h_{\text{elevation}} + h_{\text{pressure}} + h_{\text{friction}} \tag{2}$$

- **Elevation head:** assumed $h_{\text{elevation}} = 0.5\,\text{m}$.
- **Pressure head:** $\Delta P = P_2 - P_1, \quad h_{\text{pressure}} = \dfrac{\Delta P}{\rho g}$
- **Friction head:**
  (a) Reynolds number for flow behaviour, using the velocity in the chosen pipe: $\text{Re} = \dfrac{\rho\,u_\infty\,D_{\text{inner}}}{\mu}$
  (b) Fanning friction factor for turbulent flow, with $\varepsilon = 0.046\,\text{mm}$ (commercial steel; conservative for SS 316):
  $$\frac{1}{f^{1/2}} = -4\log_{10}\left[\frac{\varepsilon}{3.7\,D_{\text{inner}}} + \left(\frac{6.81}{\text{Re}}\right)^{0.9}\right]$$
  (c) Frictional energy loss per unit mass, with L the total pipe length (including vertical runs):
  $$e_f = \frac{2fLu_\infty^2}{D_{\text{inner}}}, \qquad e_f = g\,h_{\text{friction}} \;\Rightarrow\; h_{\text{friction}} = \frac{e_f}{g}$$

4. Motor power

$$P_{\text{motor}} = \frac{\dot{Q}\,\rho\,g\,h_{\text{total}}}{\eta_{\text{pump}}}, \qquad \text{Motor size} = 1.2\,P_{\text{motor}} \;\; (20\,\%\text{ capacity margin}) \tag{3}$$

5. Net positive suction head (NPSH)

An important check to prevent boiling and cavitation at the pump inlet:

$$\mathrm{NPSH}_A > \mathrm{NPSH}_R \tag{4}$$

Measured from the liquid surface in the suction tank:

$$\mathrm{NPSH}_A = \underbrace{\frac{P_{\mathrm{tank}}}{\rho g}}_{\text{tank head}} + \underbrace{h_{\mathrm{elevation}}}_{\text{elevation head}} - h_{\mathrm{friction,\ suction}} - \underbrace{\frac{P_{\mathrm{vap}}}{\rho g}}_{\text{vapour head}} \tag{5}$$

Equivalently, at the pump suction flange: $\mathrm{NPSH}_A = \dfrac{P_{\mathrm{suction}}}{\rho g} + \underbrace{\dfrac{u^2}{2g}}_{\text{velocity head}} - \dfrac{P_{\mathrm{vap}}}{\rho g}$, with $h_{\mathrm{elevation}} = 0.5\,\mathrm{m}$.

Part B: Preliminary sizing (hydraulic power only)

Assumptions

- Head $h = 7\,\text{m}$: height difference between the top and bottom of the holding tank.
- Densities from the stream table: $1005\,\text{kg/m}^3$ (media), $1097\,\text{kg/m}^3$ (glucose feed). Since $\dot{Q}\rho = \dot{m}$, density cancels out of the fluid power.
- Pump efficiency $\eta_{\text{pump}} = 0.88$.
- Friction and pressure heads are left for the full sizing (Part A).

Equations

Fluid power (also called hydraulic or water power) and shaft power:

$$P_{\text{fluid}} = \dot{Q}\,\rho\,g\,h = \dot{m}\,g\,h, \qquad P_{\text{shaft}} = \frac{P_{\text{fluid}}}{\eta_{\text{pump}}} \tag{6}$$

P-102: sterile media transfer, stream 110 (2 h per batch)

$$\dot{m} = 7196.8\,\text{kg/h} \times \frac{1\,\text{h}}{3600\,\text{s}} = 1.999\,\text{kg/s}, \qquad \dot{Q} = \frac{\dot{m}}{\rho} = \frac{1.999\,\text{kg/s}}{1005\,\text{kg/m}^3} = 1.99 \times 10^{-3}\,\text{m}^3\text{/s}\ (7.16\,\text{m}^3\text{/h})$$

$$P_{\text{fluid}} = \dot{m}\,g\,h = (1.999\,\text{kg/s})(9.81\,\text{m/s}^2)(7\,\text{m}) = 137.3\,\text{W} = 0.137\,\text{kW}$$

$$P_{\text{shaft}} = \frac{0.1373\,\text{kW}}{0.88} = \boxed{0.156\,\text{kW}}$$

P-103: sterile glucose feed, stream 112 (21 310 kg per batch, averaged over 36 h)

$$\dot{m} = \frac{21\,310\,\text{kg}}{36\,\text{h}} \times \frac{1\,\text{h}}{3600\,\text{s}} = 0.1644\,\text{kg/s}, \qquad \dot{Q} = \frac{0.1644\,\text{kg/s}}{1097\,\text{kg/m}^3} = 1.50 \times 10^{-4}\,\text{m}^3\text{/s}\ (0.54\,\text{m}^3\text{/h})$$

$$P_{\text{fluid}} = \dot{m}\,g\,h = (0.1644\,\text{kg/s})(9.81\,\text{m/s}^2)(7\,\text{m}) = 11.3\,\text{W} = 0.0113\,\text{kW}$$

$$P_{\text{shaft}} = \frac{0.0113\,\text{kW}}{0.88} = \boxed{0.0128\,\text{kW}}$$

The glucose is sterilized in about 3 h and held in T-103; P-103 then feeds the fermenter from storage across the fed-batch phases, so the 36 h average sets its preliminary duty.

Pump type

Most pumps in the plant are positive-displacement, since each moves a known, fixed quantity of fluid:

- **P-102:** positive-displacement; transfers a set batch of sterile media, so only the head matters.
- **P-103:** positive-displacement; meters a known quantity of sterile glucose to the fermenter.
- **Exception, P-304** (immediately downstream of the bioreactor): centrifugal. It recycles fluid through the filtration unit back to the reactor at a variable rate set by differential pressure, which a positive-displacement pump would handle poorly.